



Double Sideband Suppressed Carrier Systems(DSBSC)

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Need of DSBSC

- If $P_c=100\text{watts}$, $m=0.5$

$$P_t = P_c \left(1 + \frac{m^2}{2}\right)$$

- Total power= 112.5 watts

$$P_{SB} = P_t - P_c$$

- Total sideband power= 12.5 watts

$$\rightarrow \omega_c + \omega_m \text{ (USB) \& } (\omega_c - \omega_m) \text{ (LSB)}$$

- Carrier does not contain any information

- Waste of power by transmitting carrier signal

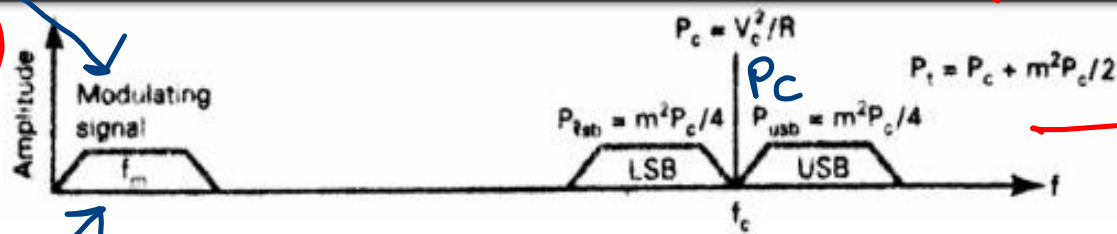
Need and Principle of DSBSC

- Conventional AM (DSBFC) requires more bandwidth.
- In DSBFC wave more power is consumed in carrier. It does not carry any information.
- Sideband carries same information.

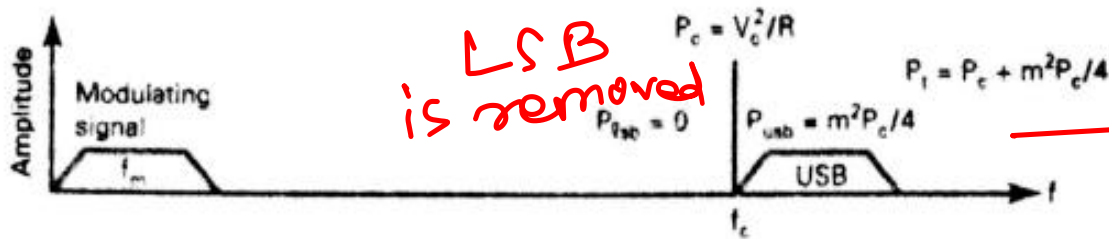
Modulation Index (m)	Carrier Power (W)	Total Power (W)	Sideband Power (W)	Efficiency (%)
1.5 (High Modulation)	200 W	425 W	225 W	52.9%
0.2 (Low Modulation)	50 W	51 W	1 W	2.0%
0 (No Modulation)	100 W	100 W	0 W	0%

Other Variants of AM (frequency spectrum)

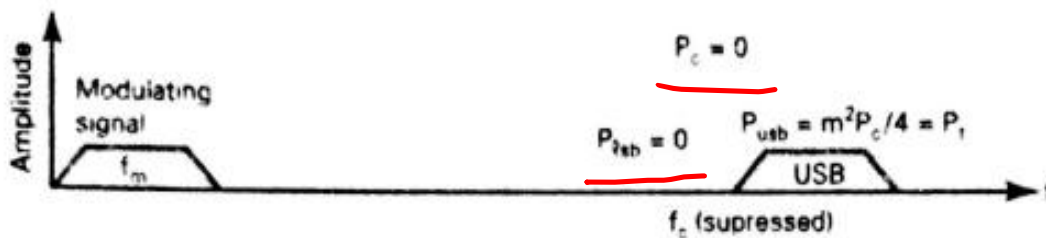
Amplitude
Frequency
f



DSBFC



SSB

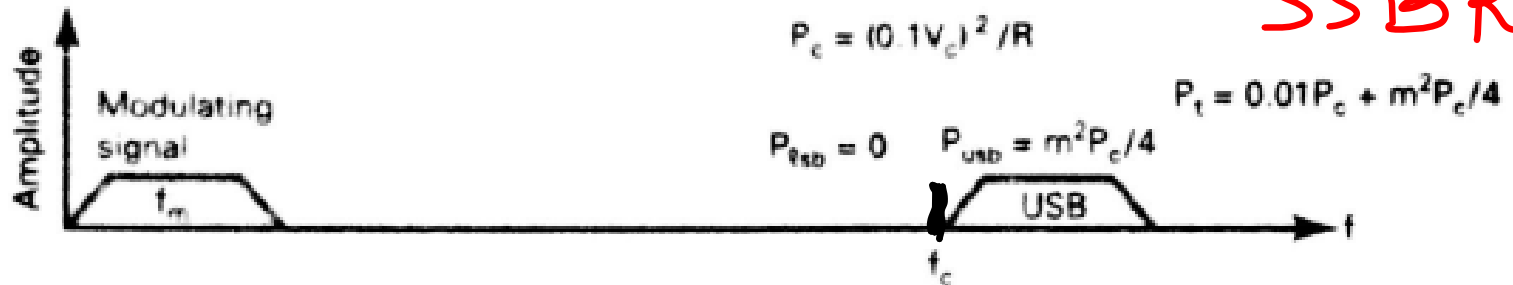


SSBSC

(c)

Other Variants of AM

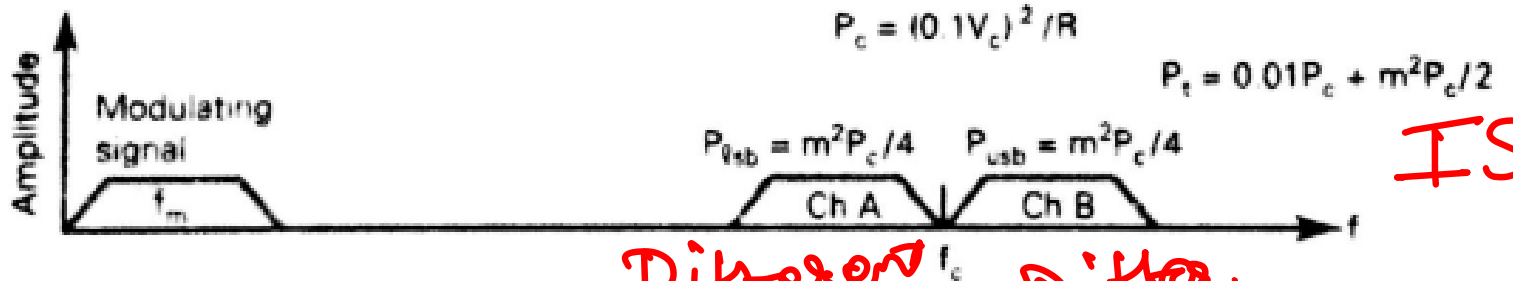
4



SSBRC

(d)

5

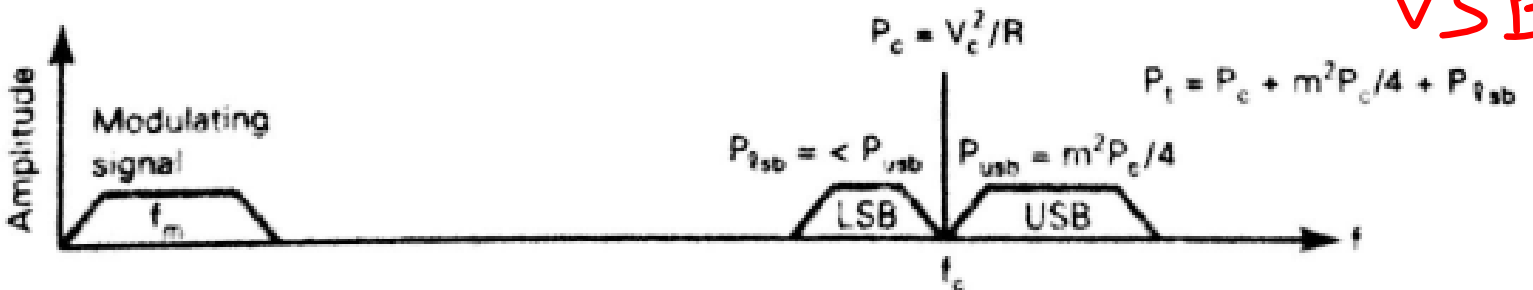


ISB

Different Different.

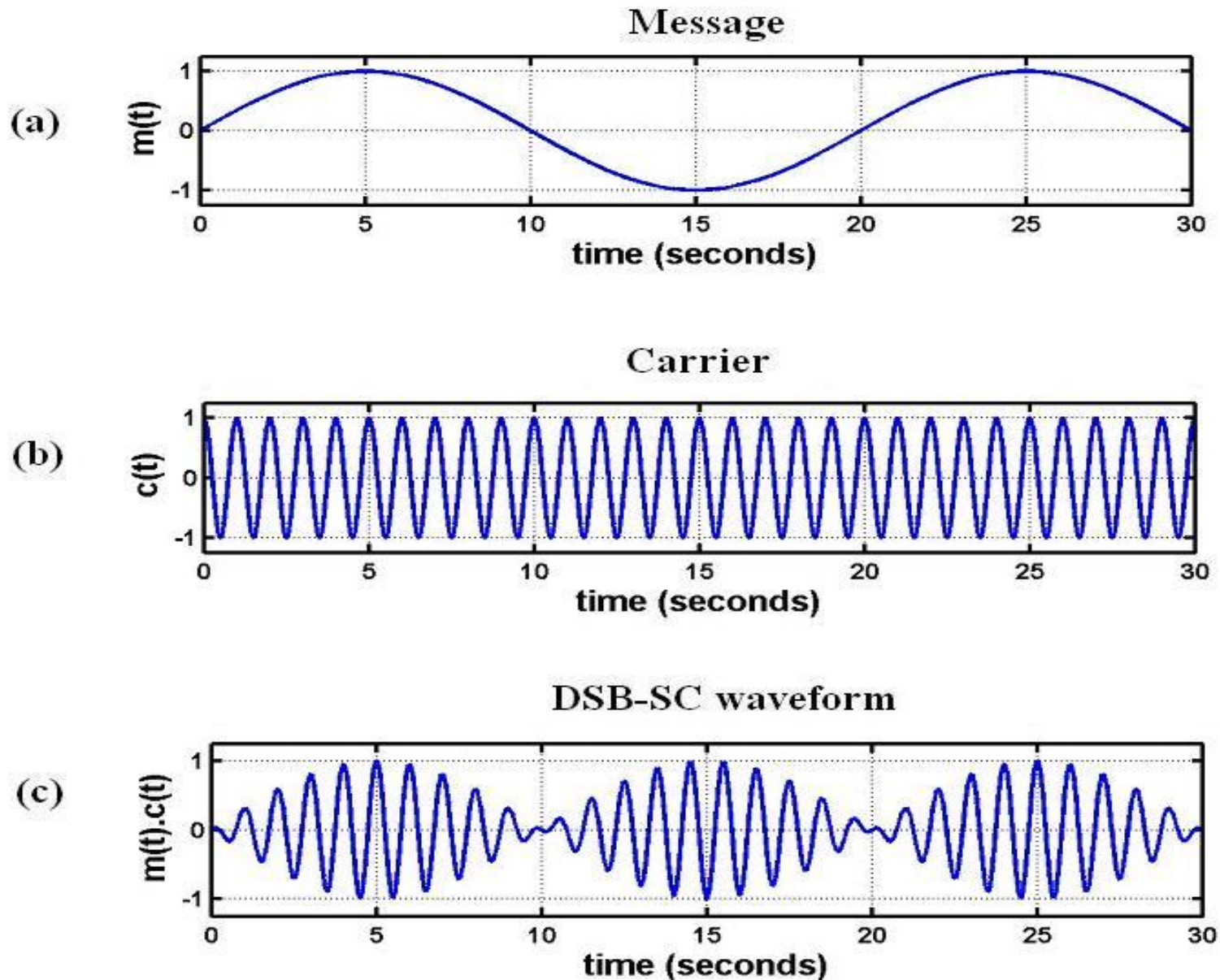
(e)

6

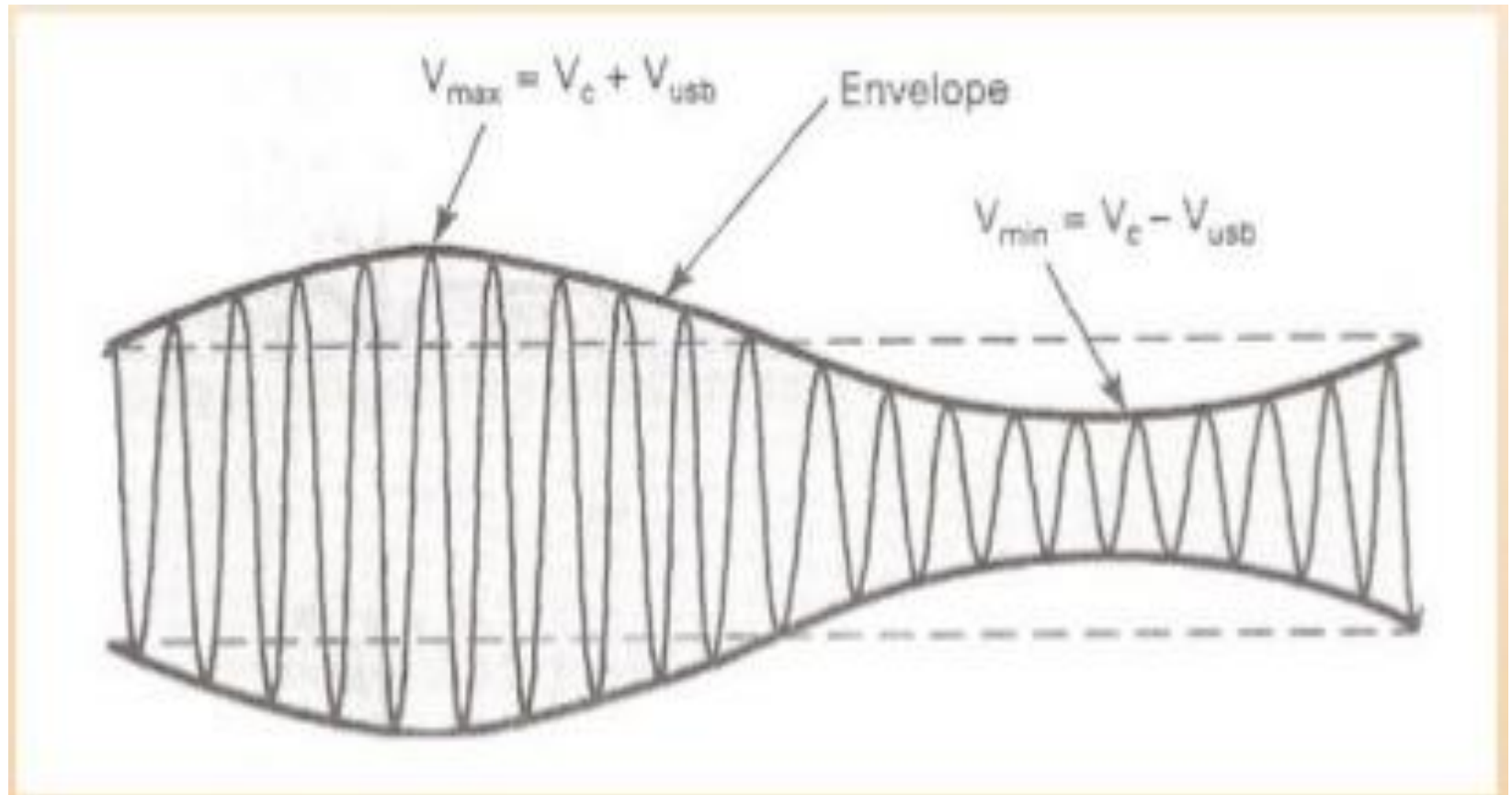


VSB

DSBSC Wave

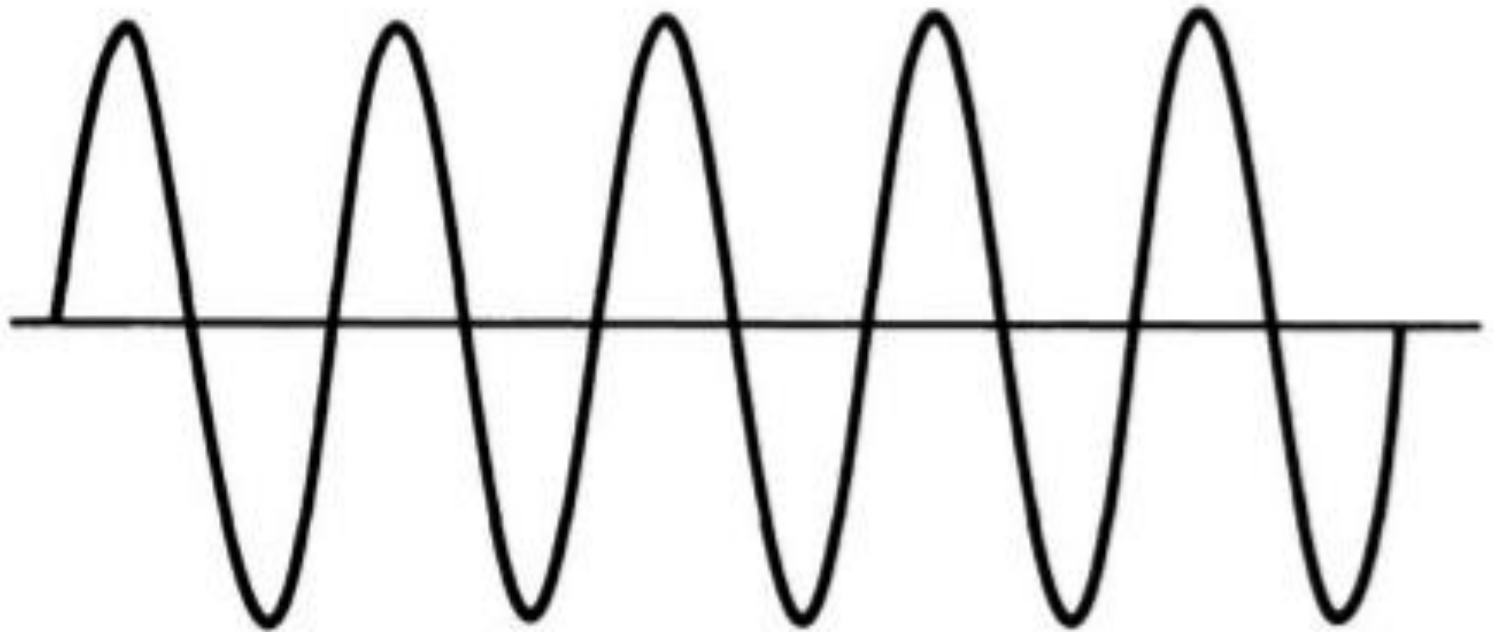


SSBFC Wave

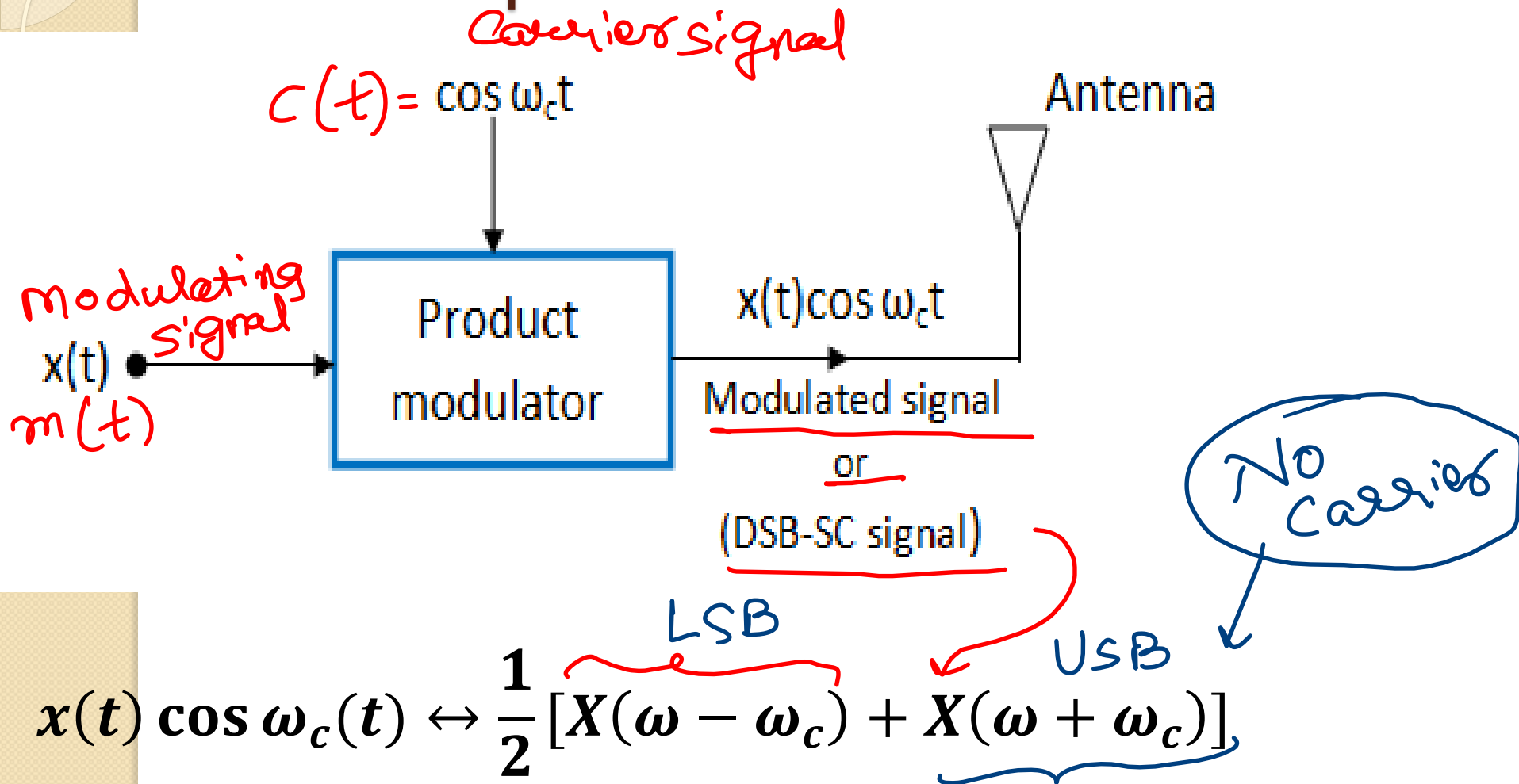


SSBSC Wave

$$V_p = V_{usf}$$



Basic Principle of DSBSC wave



Basic Principle of DSBSC wave

$$V_{am}(t) = [(1 + m \sin(2\pi f_m t))][V_c \sin(2\pi f_c t)]$$

Expression
of AM
DSBFC

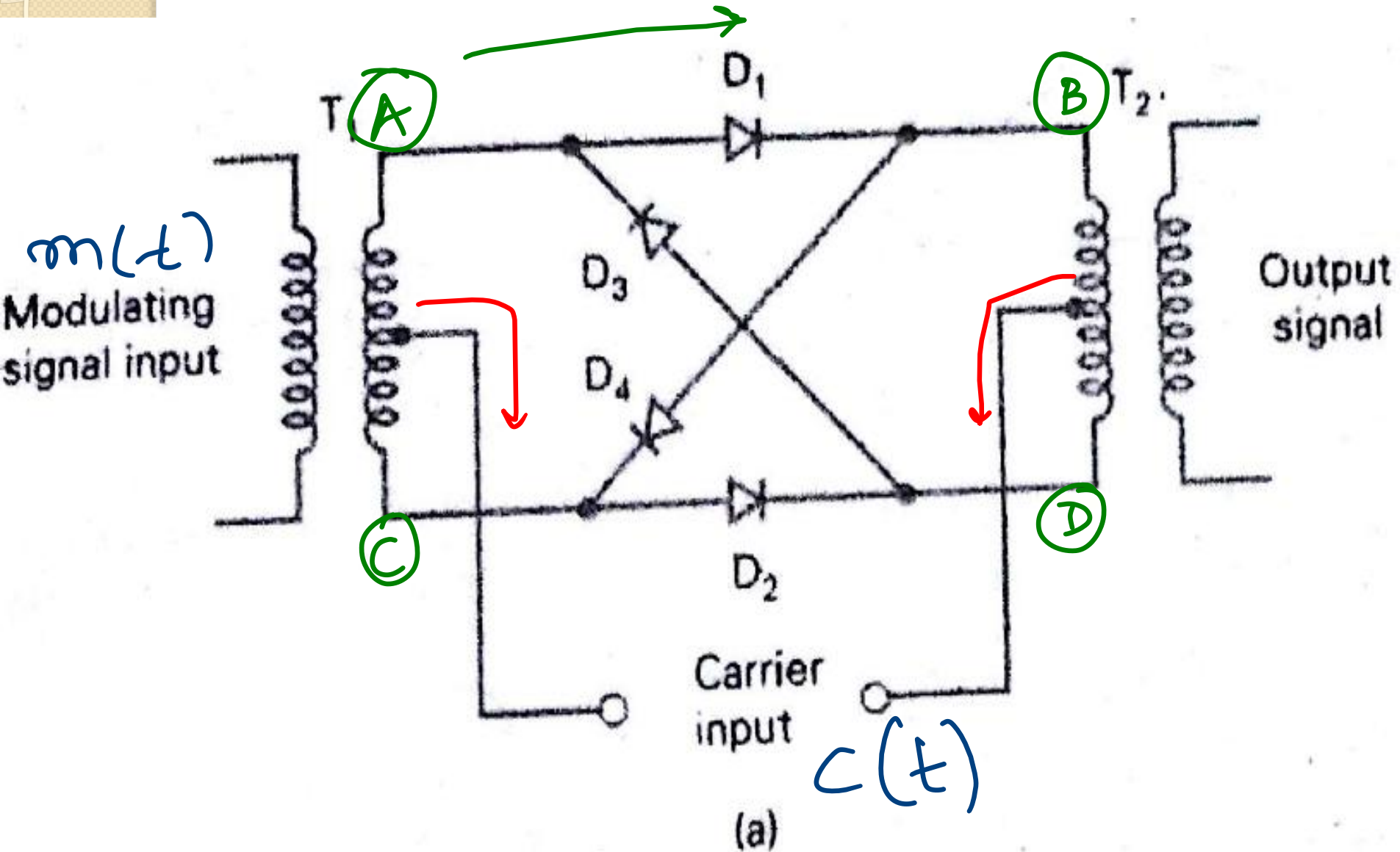
- **If constant component is removed from the modulating signal then,**

$$V_{am}(t) = [m \sin(2\pi f_m t)][V_c \sin(2\pi f_c t)]$$

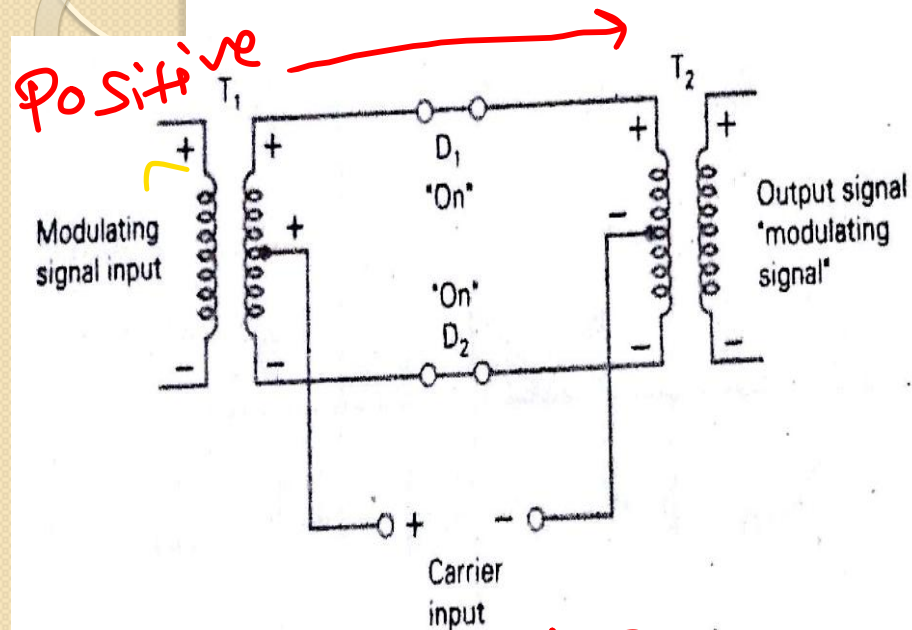
$$V_{am}(t) = \frac{mV_c}{2} \cos[2\pi(f_c - f_m)t] - \frac{mV_c}{2} \cos[2\pi(f_c + f_m)t]$$

DSBSC

DSBSC Generation- Ring Modulator

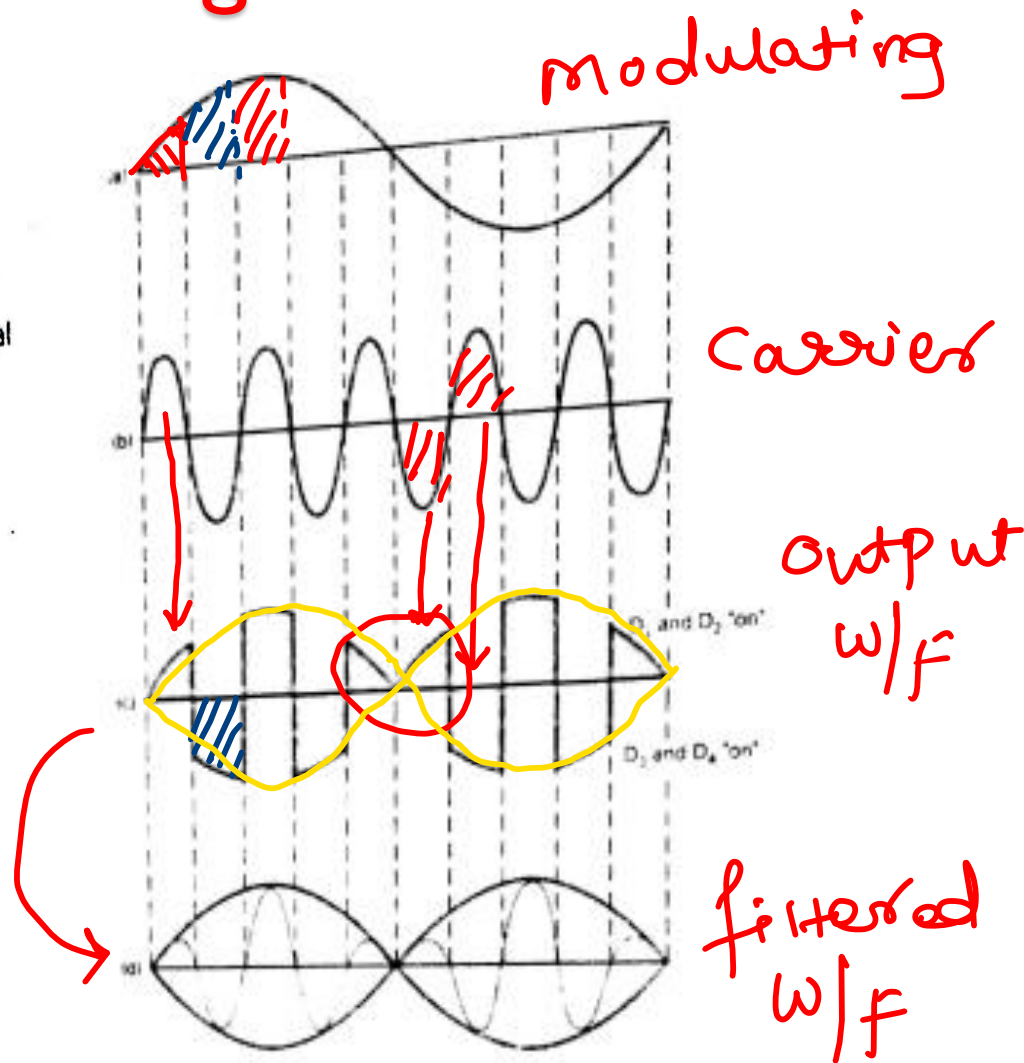


DSBSC Generation- Ring Modulator

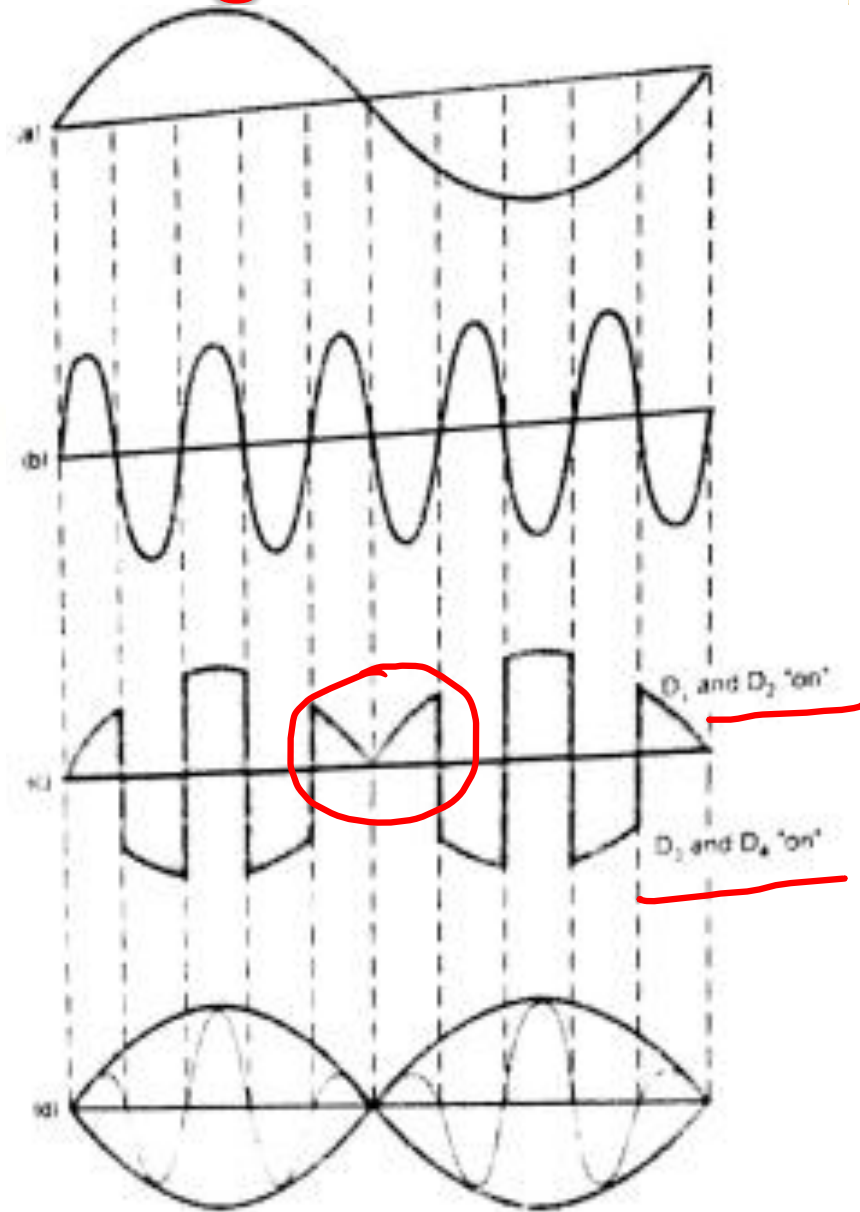
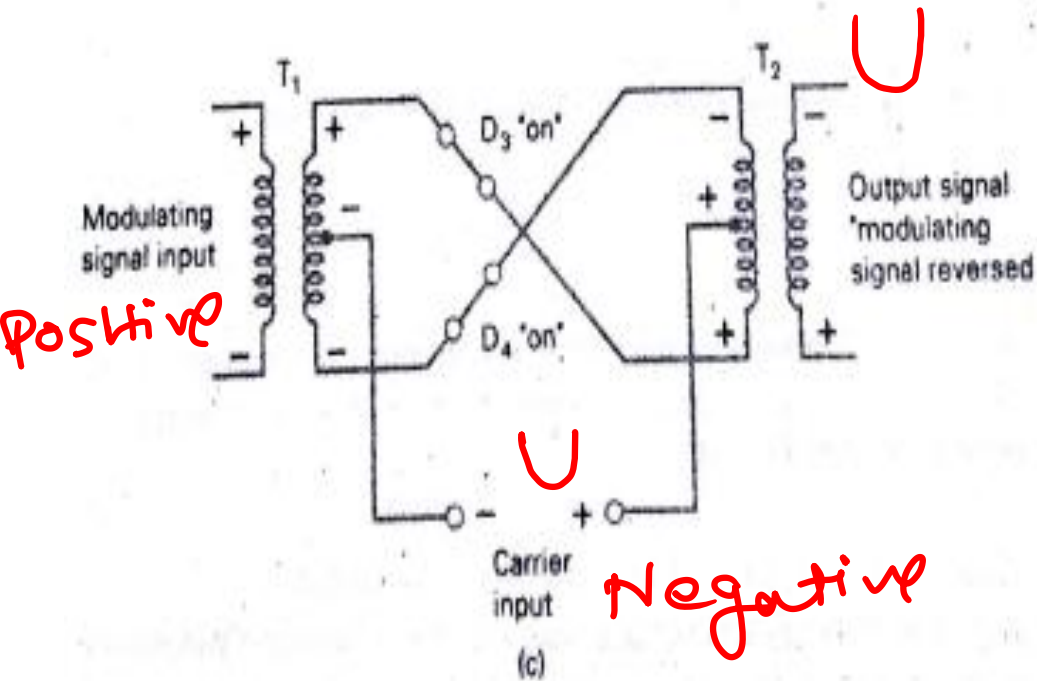


$m(t)$	$c(t)$	$m(t)$	$c(t)$
-ve	+ve	+ve	+ve
-ve	-ve	+ve	-ve

Positive



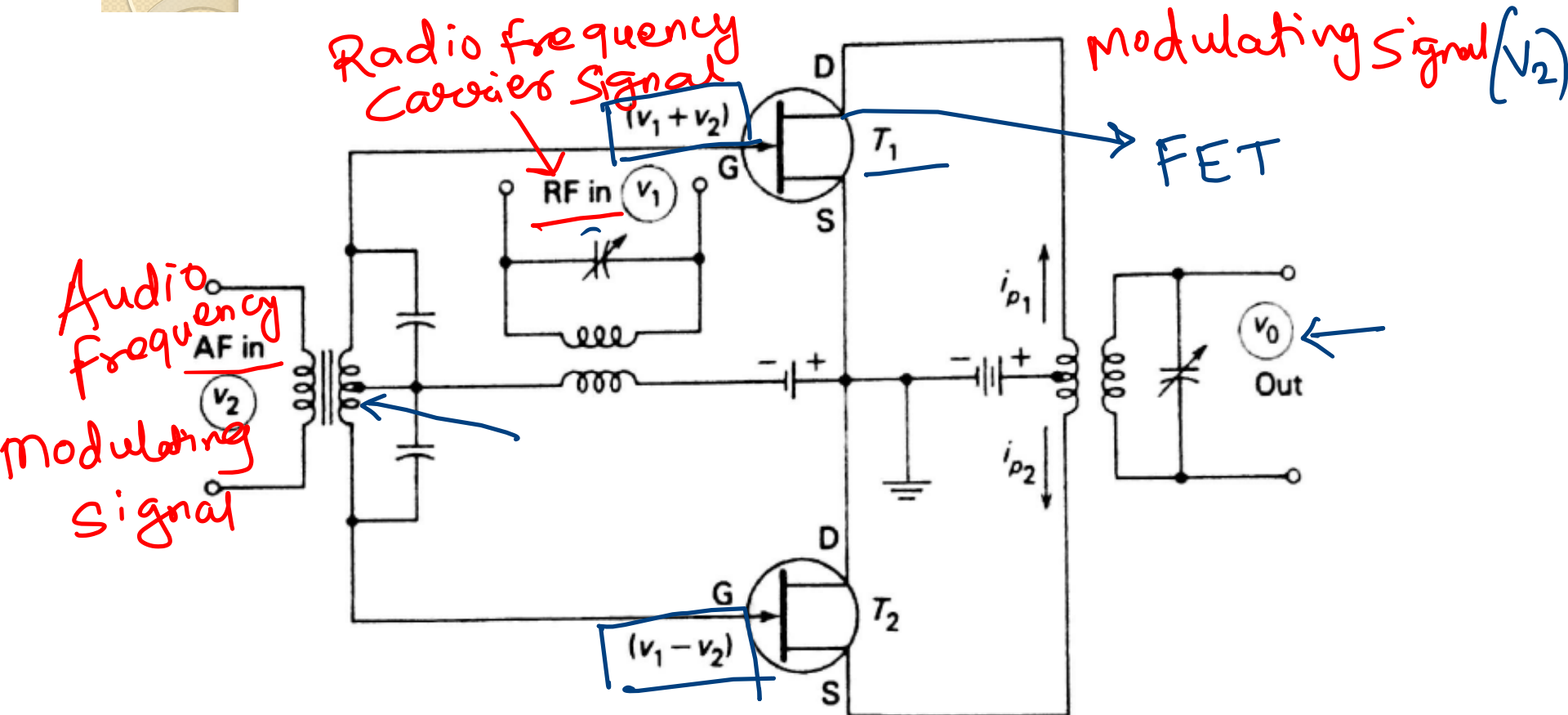
DSBSC Generation- Ring Modulator



2

DSBSC Generation- Balanced

Modulator



the drain current through FET is given by

$$i_d = a + b x + c x^2 \rightarrow x \rightarrow \text{Input}$$

$$x = v_1 + v_2$$

$$\begin{aligned} i_{d1} &= a + b(v_1 + v_2) + c(v_1 + v_2)^2 \\ &= a + b v_1 + b v_2 + c v_1^2 + c v_2^2 + 2c v_1 v_2 \quad \text{--- (1)} \end{aligned}$$

$$\begin{aligned} i_{d2} &= a + b(v_1 - v_2) + c(v_1 - v_2)^2 \\ &= a + b v_1 - b v_2 + c v_1^2 + c v_2^2 - 2c v_1 v_2 \quad \text{--- (2)} \end{aligned}$$

$$i_1 = i_{d1} - i_{d2} = 2 b v_2 + 4 c v_1 v_2 \quad \text{--- (3)}$$

$$\left. \begin{array}{l} V_1 \rightarrow \text{Carrier signal} \rightarrow V_c \sin \omega_c t \\ V_2 \rightarrow \text{Modulating s/g} \rightarrow V_m \sin \omega_m t \end{array} \right\} \text{Substitute in eqn (3)}$$

$$i_1 = 2b(V_m \sin \omega_m t) + 4c(V_c \sin \omega_c t)(V_m \sin \omega_m t)$$

$$\boxed{\sin A \cdot \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]}$$

$$i_1 = 2bV_m \sin \omega_m t + \cancel{4}cV_cV_m \frac{1}{2} [\cos(\omega_c - \omega_m) - \cos(\omega_c + \omega_m)]$$

$$V_o \propto i_1 \Rightarrow V_o = \propto i_1$$

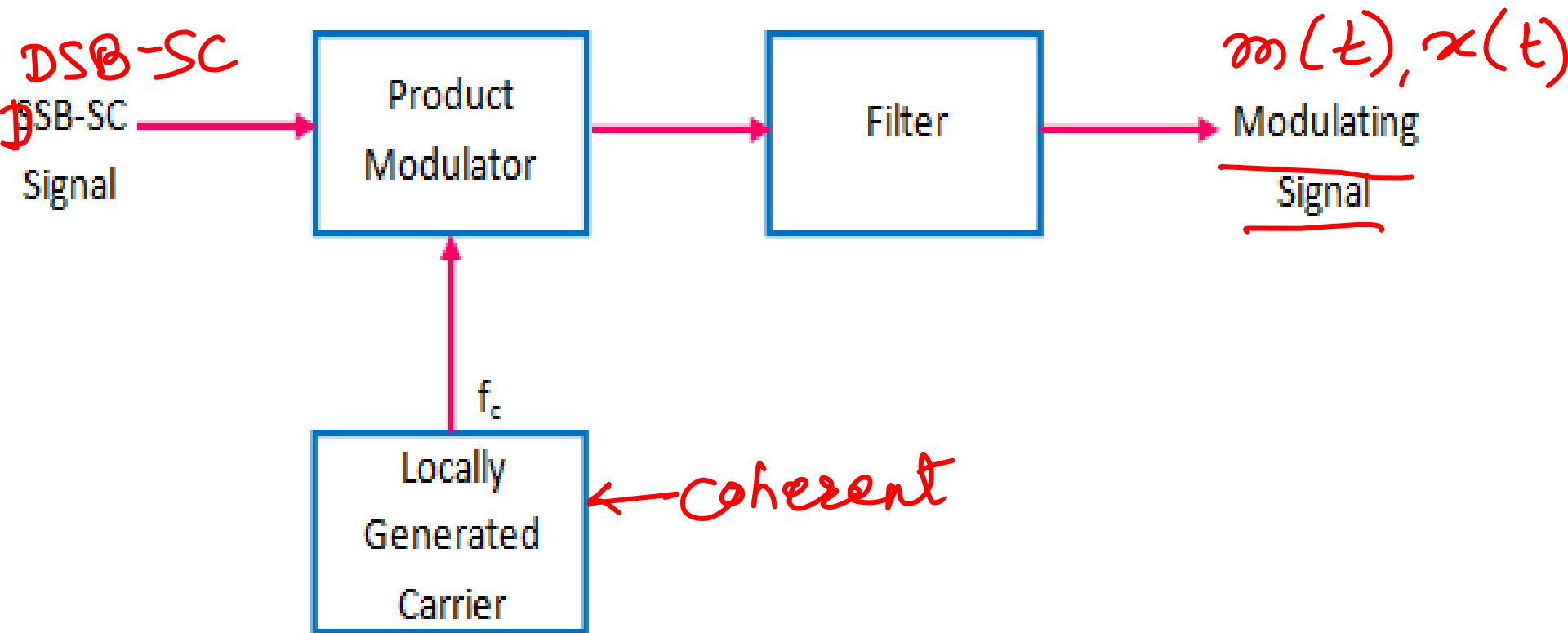
$$V_o = 2\propto bV_m \sin \omega_m t + \cancel{2}cV_cV_m [\cos(\omega_c - \omega_m) - \cos(\omega_c + \omega_m)]$$

$$V_o = \underbrace{P \sin \omega_m t}_{\text{Modulating Signal}} + Q \left[\underbrace{\cos(\omega_c - \omega_m)}_{\text{Lower Sideband}} - \underbrace{\cos(\omega_c + \omega_m)}_{\text{Upper Sideband}} \right]$$

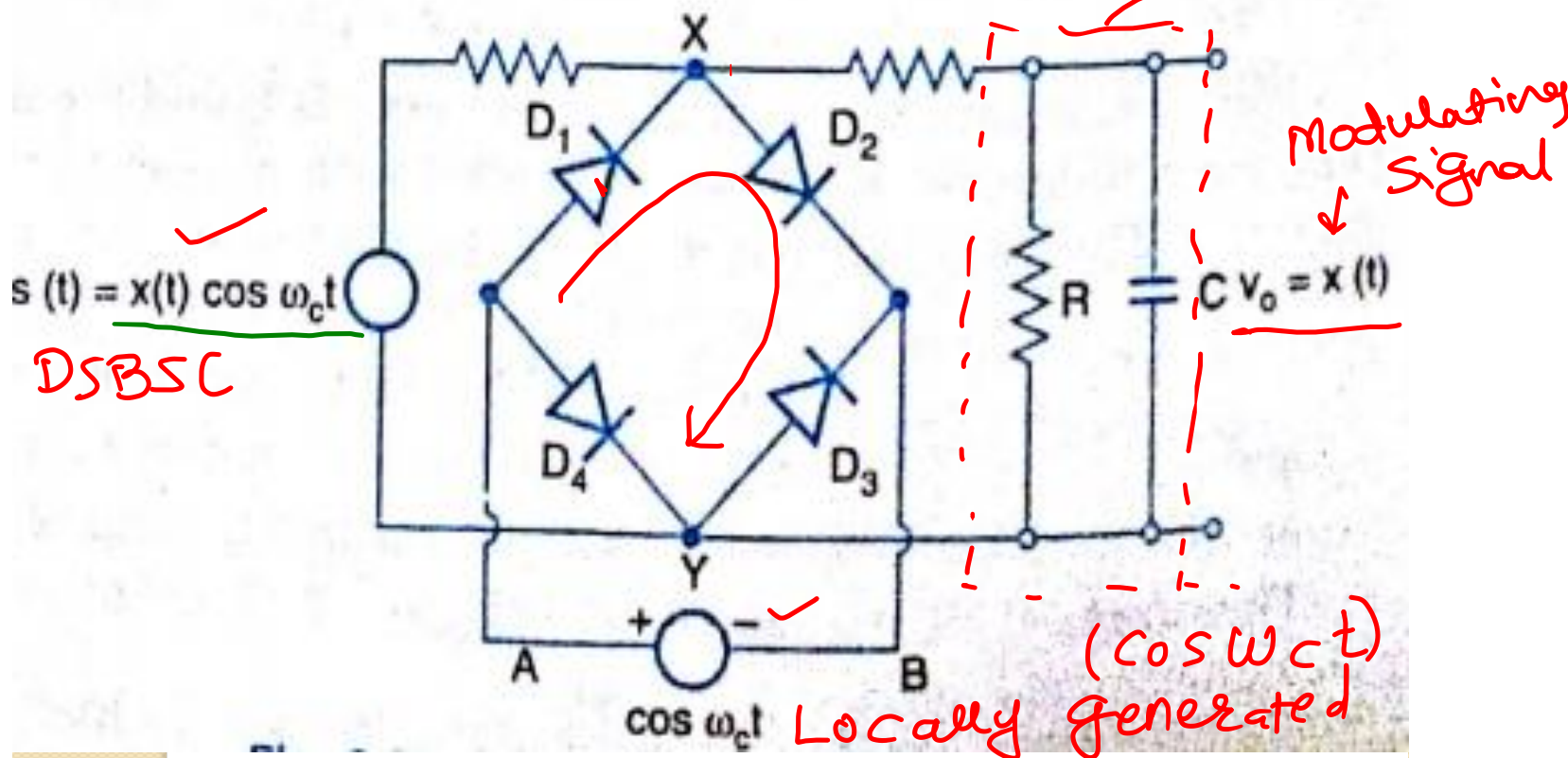
~~X~~ No Carrier

DSBSC Signal

Coherent DSBSC Detection (Demodulation)



DSBSC Demodulation by switching (chopper) demodulator



The output voltage is given by

$$V_o = S(t) * \cos \omega_c t$$

↳ DSBSC signal at Input

Locally generated carrier

$$V_o = x(t) \cos \omega_c t \cdot \cos \omega_c t$$

$$= x(t) \cos^2 \omega_c t$$

$$= x(t) \left[\frac{1 + \cos 2\omega_c t}{2} \right]$$

$$V_o(t) = \frac{x(t)}{2} + \frac{1}{2} \cos(2\omega_c t) \cdot x(t)$$

Removed by filter

Scaled version of modulating signal

DSBSC Demodulation by switching (chopper) demodulator

